

GSC-X Reliability Sandbox

Reliability Validation V1: Multi-Dataset Validation of Predictive Health Monitoring, Early Warning and Prognostics Using Context Memory and Early Limit Indicator

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GSC-X Repository
<https://github.com/sisacek66198068/GSC-X>

Related GSC-X Sandbox Portfolio

GSC-X Industrial Sandbox
GSC-X Reliability Sandbox
GSC-X Fusion Sandbox
GSC-X Spaceflight Sandbox
GSC-X Battery Sandbox
GSC-X Robotics Sandbox
GSC-X Nuclear Sandbox
GSC-X Energy Network Sandbox

Patent-Pending Architecture Portfolio

PV 2026-216 (GSC-F)
PV 2026-236 (GSC-S)
PV 2026-241 (GSC-C)
PV 2026-242 (GSC-X)

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Abstract

GSC-X Reliability Sandbox represents the predictive health-monitoring and early-warning branch of the broader GSC-X validation ecosystem.

This work presents Reliability Validation V1, a multi-dataset validation campaign designed to evaluate predictive warning, degradation tracking, prognostics and health monitoring capabilities using the executive-state architecture of GSC-X.

The validation portfolio includes NASA CMAPSS turbofan degradation datasets, IMS Bearing datasets and the CWRU bearing fault benchmark.

Across NASA CMAPSS FD001–FD004, the validation campaign covered 709 turbofan engines and achieved 100% detection of degradation trajectories.

The strongest CM+ELI configuration achieved a median Warning Quality Score (WQS) approaching 0.9 while maintaining complete detection coverage.

Additional validation using IMS Bearings demonstrated a lead-horizon improvement from approximately 0.65% to 13.64%, while CWRU Robust V2 achieved perfect separation between normal and fault states with an F1 score of 1.00.

The primary contribution of this work is not a new predictive-maintenance algorithm alone.

The broader contribution is the demonstration that Context Memory (CM) and Early Limit Indicator (ELI) may provide a common predictive-awareness framework across multiple reliability domains.

Together with previously completed Battery Validation V1 and Energy Validation V1, the Reliability Sandbox contributes to an emerging predictive-resilience framework spanning asset health monitoring, energy storage systems and critical infrastructure protection.

The presented results should be interpreted as validation evidence obtained on public benchmark datasets rather than proof of operational deployment performance.

Nevertheless, the consistency of results across multiple independent datasets provides encouraging evidence that executive-state architectures may represent a promising direction for future predictive-maintenance and infrastructure-resilience technologies.

1 Introduction

Modern industrial infrastructure increasingly depends on the ability to detect degradation before failures occur.

Turbofan engines, rotating machinery, industrial equipment, energy systems and critical infrastructure assets all exhibit complex degradation trajectories that evolve over time.

Traditional monitoring systems are typically reactive.

Warnings are often generated only after threshold violations or significant performance loss has already occurred.

As infrastructure systems become larger, more interconnected and more economically important, the value of predictive awareness continues to increase.

The GSC-X Reliability Sandbox investigates a complementary approach.

Rather than focusing exclusively on fault detection, the Reliability Sandbox evaluates whether executive-state variables can provide meaningful predictive warning before conventional failure boundaries are reached.

The architecture is built around two executive-state variables:

- Context Memory (CM)
- Early Limit Indicator (ELI)

Together these variables attempt to capture accumulated system context, degradation persistence, recovery dynamics and approach toward instability boundaries.

The objective is not to replace existing predictive-maintenance systems.

Instead, GSC-X is evaluated as an additional executive layer capable of extending operational awareness through predictive warning and degradation tracking.

Reliability Validation V1 forms part of a broader validation ecosystem.

Previous validation campaigns have been completed within Battery Sandbox and Energy Network Sandbox environments.

Together these validation branches explore whether common executive principles can support predictive resilience across multiple classes of critical infrastructure.

The present work focuses specifically on reliability engineering, predictive maintenance and degradation forecasting using public benchmark datasets.

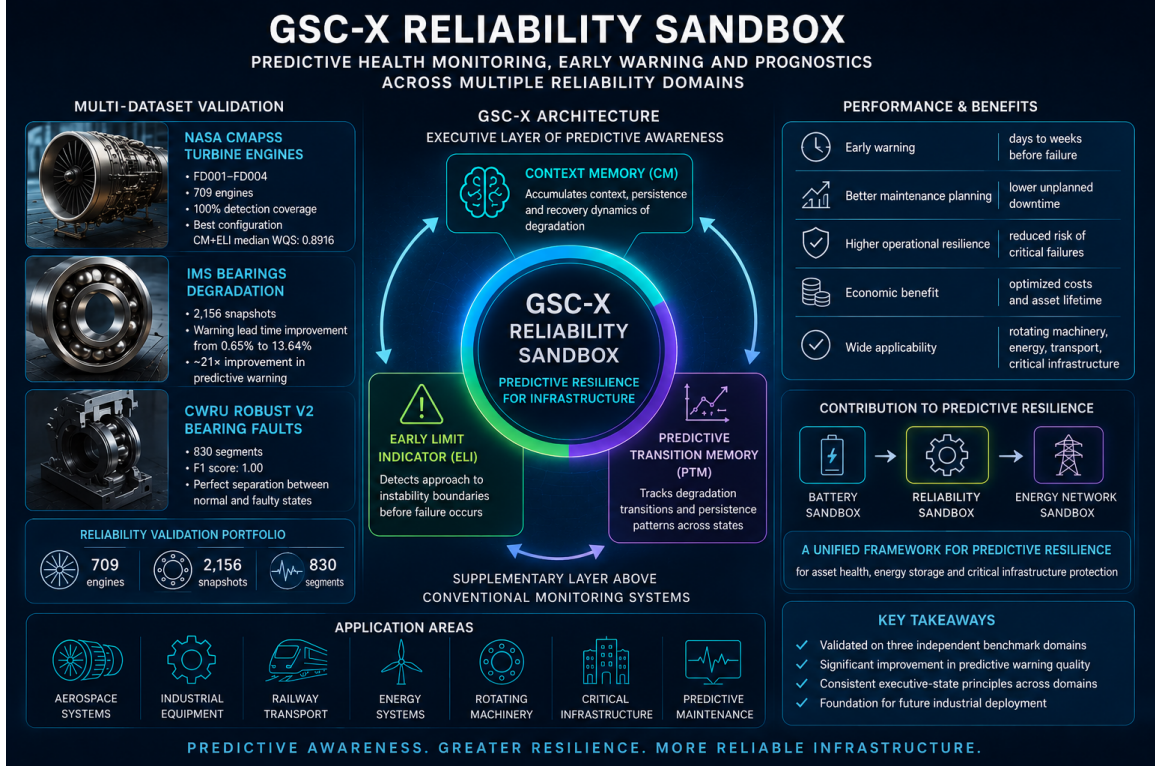


Figure 1: Conceptual overview of the GSC-X Reliability Sandbox. The diagram summarizes the validation portfolio, executive-state architecture (CM, ELI, PTM), application domains and the broader predictive-resilience ecosystem investigated in this work.

2 GSC-X Reliability Architecture

The Reliability Sandbox evaluates GSC-X as an executive predictive-awareness layer operating above conventional monitoring systems.

The architecture is based on three primary executive-state components:

- Context Memory (CM)
- Early Limit Indicator (ELI)
- Predictive Transition Memory (PTM)

Context Memory accumulates operational history and degradation context.

Rather than evaluating system state solely from instantaneous measurements, CM incorporates persistence, recovery dynamics and accumulated degradation behavior.

Early Limit Indicator is designed to identify approach toward instability boundaries before conventional threshold violations occur.

Instead of reacting to failure events, ELI attempts to provide anticipatory awareness of emerging degradation regimes.

Predictive Transition Memory extends this concept by tracking degradation transitions and persistence patterns across operational states.

PTM is intended to capture the dynamic evolution of degradation trajectories rather than isolated measurements.

Together these executive-state variables form a predictive layer designed to operate as a complementary system above conventional monitoring architectures.

The Reliability Sandbox investigates whether these variables can improve predictive awareness across multiple classes of reliability datasets.

The architecture is intentionally designed as a platform-independent executive layer.

No assumptions are made regarding specific sensors, hardware vendors or industrial environments.

The objective is to evaluate predictive behavior rather than implementation-specific performance.

3 Validation Portfolio

Reliability Validation V1 was designed as a multi-dataset validation campaign.

Rather than optimizing performance on a single benchmark, the objective was to evaluate whether predictive-warning behavior remains reproducible across independent datasets representing different degradation mechanisms and machine classes.

The validation portfolio currently includes the datasets summarized in Table 1.

Table 1: Reliability Validation Portfolio

Dataset	Scale	Status
NASA CMAPSS FD001–FD004	709 engines	Completed
IMS Bearings 1st Test	2156 snapshots	Completed
CWRU Bearings Robust V2	830 segments	Completed

The selected datasets span multiple reliability domains including turbofan engines, rotating machinery and bearing fault diagnostics.

This diversity allows evaluation of predictive-warning behavior under substantially different degradation conditions.

The primary objective of Reliability Validation V1 is therefore reproducibility across datasets rather than optimization for a single benchmark environment.

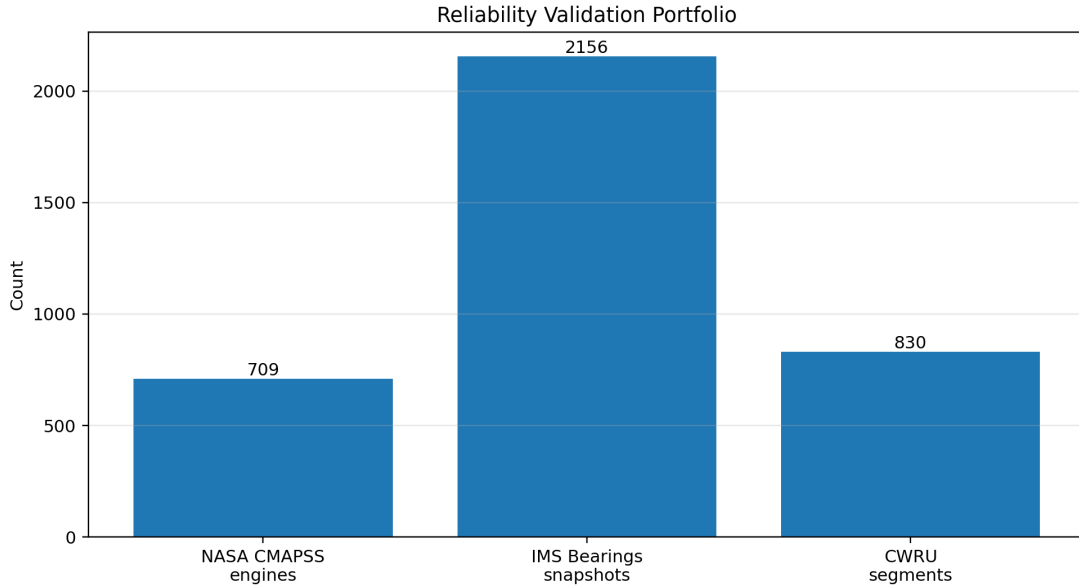


Figure 2: Scale of the primary validation datasets used in Reliability Validation V1. The validation portfolio spans turbofan-engine degradation (NASA CMAPSS), bearing degradation telemetry (IMS) and machine-fault diagnostics (CWRU), providing coverage across multiple independent reliability domains.

4 NASA CMAPSS Validation

The NASA Commercial Modular Aero-Propulsion System Simulation (CMAPSS) dataset represents the primary validation benchmark used within Reliability Validation V1.

NASA CMAPSS is widely used for predictive-maintenance research and contains degradation trajectories of turbofan engines operating under varying conditions.

The validation campaign included all four major dataset groups:

- FD001
- FD002
- FD003
- FD004

The complete campaign therefore covered 709 individual engines.

The objective was not merely failure detection.

Instead, the focus was placed on predictive-warning quality and the ability to identify degradation trajectories before conventional failure boundaries were reached.

The baseline configuration produced no meaningful predictive warning quality.

The addition of Context Memory improved predictive performance substantially.

The strongest results were obtained when Context Memory and Early Limit Indicator were combined into a unified executive-state architecture.

The resulting validation metrics are summarized in Table 2.

Table 2: NASA CMAPSS Validation Summary

Metric	Result
Dataset groups	FD001–FD004
Total engines	709
Detection coverage	100%
Baseline median WQS	0.0000
CM median WQS	0.2319
CM+ELI median WQS	0.8916

The NASA campaign therefore represents the largest validation component of Reliability Validation V1.

The results suggest that executive-state variables may significantly improve predictive awareness compared to threshold-based monitoring alone.

5 NASA Reliability Results

The NASA validation campaign provides the strongest quantitative evidence currently available within the Reliability Sandbox.

Several observations are noteworthy.

First, complete detection coverage was maintained across all 709 engines.

Second, the progression from Baseline to CM and subsequently to CM+ELI demonstrates a consistent increase in predictive-warning quality.

Third, the improvement was observed across multiple dataset groups rather than being restricted to a single operating condition.

This behavior is important because predictive-maintenance systems frequently exhibit strong performance only under narrow benchmark configurations.

The NASA results indicate that executive-state variables remain useful across broader degradation conditions.

The validation therefore supports the hypothesis that predictive warning can emerge from accumulated contextual awareness rather than from threshold detection alone.

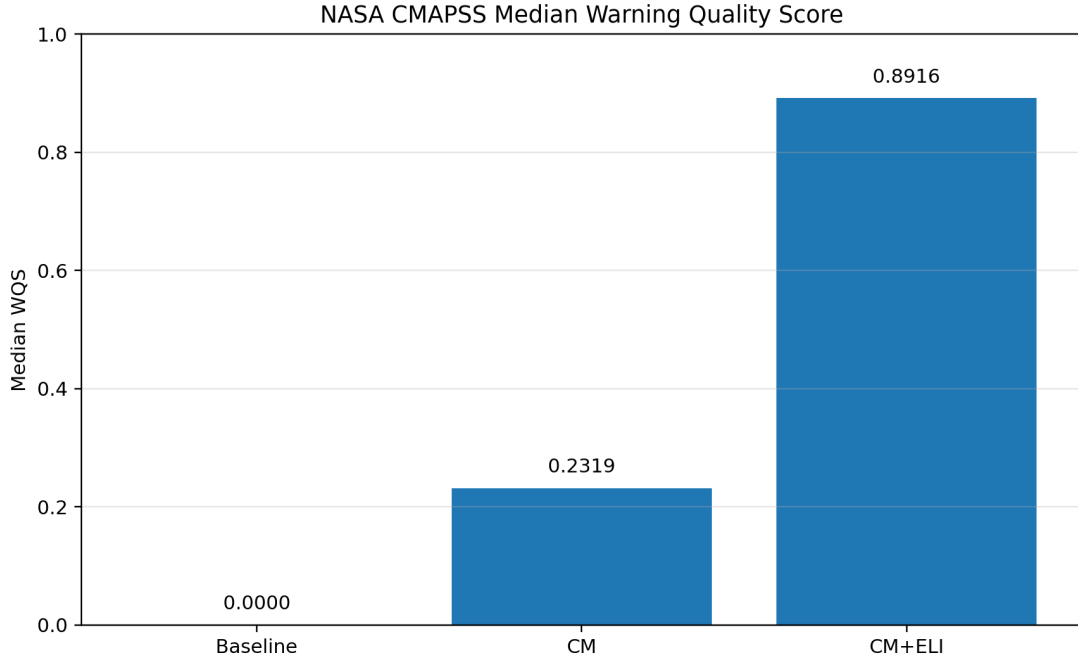


Figure 3: Median Warning Quality Score (WQS) progression during NASA CMAPSS validation. The executive-state architecture based on Context Memory (CM) and Early Limit Indicator (ELI) substantially improved predictive-warning quality compared to baseline monitoring.

6 IMS Bearing Validation

To complement the NASA turbofan benchmark, Reliability Validation V1 also evaluated predictive-warning behavior using the IMS Bearing Dataset.

The IMS dataset contains long-duration vibration measurements collected from rolling-element bearings operating under controlled degradation conditions.

Unlike NASA CMAPSS, which focuses on engine degradation trajectories, IMS provides a reliability benchmark based on real vibration telemetry and progressive bearing wear.

The first validation cycle focused on the IMS 1st Test dataset containing 2156 time snapshots across multiple vibration channels.

An initial RMS-only implementation demonstrated limited predictive capability.

Warnings were generated very close to the final degradation stage, producing a lead horizon of approximately 0.65%.

To improve predictive awareness, the Reliability Sandbox introduced a multi-feature executive layer combining:

- RMS
- Standard deviation
- Kurtosis
- Crest factor
- Impulse factor
- Spectral energy

- Spectral peak

These features were combined with Context Memory and persistence tracking mechanisms. The resulting feature-layer validation significantly improved predictive performance.

Table 3: IMS Bearing Validation Results

Metric	Result
Dataset	IMS 1st Test
Snapshots	2156
RMS-only lead horizon	0.65%
Feature-layer lead horizon	13.64%
Improvement factor	$\sim 21\times$

The IMS validation demonstrates that predictive warning quality can be improved substantially when executive-state variables are combined with richer telemetry features.

The observed improvement from approximately 0.65% to 13.64% lead horizon represents one of the strongest reliability improvements observed within the current validation portfolio.

Importantly, the result was achieved without introducing domain-specific bearing models.

The improvement emerged primarily from executive-state awareness and persistence tracking.

7 CWRU Bearing Validation

Additional validation was performed using the Case Western Reserve University (CWRU) Bearing Dataset.

The CWRU benchmark is widely used for machine-health monitoring and bearing fault diagnosis research.

The objective of the Reliability Sandbox was not merely fault classification.

Instead, the benchmark was used to evaluate whether executive-state variables could consistently separate healthy and faulty operational regimes.

The Robust V2 validation campaign included:

- normal operating conditions,
- multiple fault severities,
- multiple fault locations,
- multiple independent datasets.

The resulting benchmark covered 830 signal segments distributed across ten fault datasets.

Table 4: CWRU Robust V2 Validation Results

Metric	Result
Segments analyzed	830
Fault datasets	10
Normal/Fault separation	Perfect
F1 Score	1.00

The Robust V2 campaign achieved complete separation between healthy and faulty bearing states.

Although the benchmark represents a classification-oriented validation environment, the results provide additional evidence that executive-state variables can maintain strong discrimination across independent machine-health datasets.

The CWRU benchmark therefore serves as an important complement to NASA CMAPSS and IMS validations.

Together, these datasets provide coverage across multiple reliability domains including turbofan degradation, bearing wear and machine-fault diagnostics.

8 Predictive Executive Layer

A central objective of Reliability Validation V1 is the evaluation of executive-state awareness rather than dataset-specific optimization.

Across NASA, IMS and CWRU validations, the same core executive principles were repeatedly applied.

These principles are summarized below.

- Context Memory (CM)
- Early Limit Indicator (ELI)
- Predictive Transition Memory (PTM)

Context Memory accumulates historical operational context and degradation persistence.

Early Limit Indicator attempts to identify approach toward instability boundaries before threshold violations occur.

Predictive Transition Memory tracks regime transitions and degradation evolution over time.

Together these variables create an executive-state representation that extends beyond instantaneous telemetry.

Rather than reacting exclusively to measured values, the architecture evaluates how the system arrived at its current state and whether observed trends suggest increasing instability.

This distinction is important.

Traditional monitoring systems often answer the question:

What is happening now?

The executive-state layer attempts to answer a complementary question:

What is likely to happen next?

Reliability Validation V1 therefore focuses on predictive awareness rather than reactive fault detection.

9 Discussion

Reliability Validation V1 demonstrates that executive-state architectures may provide meaningful predictive awareness across multiple reliability domains.

The most important observation is not the performance achieved on any individual benchmark.

Rather, the most significant result is the consistency of behavior observed across independent datasets representing fundamentally different degradation mechanisms.

NASA CMAPSS focuses on turbofan-engine degradation.

IMS Bearings focuses on progressive bearing wear and vibration telemetry.

CWRU focuses on machine-fault diagnostics and fault-state separation.

Despite these substantial differences, Context Memory and Early Limit Indicator consistently contributed to improved predictive awareness.

This observation suggests that executive-state variables may capture aspects of degradation dynamics that are not restricted to a specific machine class.

The results do not prove the existence of a universal predictive-maintenance framework.

However, they provide encouraging evidence that common executive principles may remain useful across multiple reliability domains.

From a broader perspective, Reliability Validation V1 complements previous Battery Validation V1 and Energy Validation V1 studies.

Together these validation branches investigate different manifestations of the same underlying challenge:

- degradation,
- instability,
- loss of operational margin,
- approach toward failure boundaries.

The convergence of results across these domains suggests that predictive resilience may be a useful organizing concept for future infrastructure technologies.

Reliability engineering, battery-health monitoring and energy-network stabilization are traditionally treated as separate disciplines.

The GSC-X validation ecosystem explores whether common executive-state principles may contribute to all three.

This possibility remains a research hypothesis.

Nevertheless, the consistency of observed behavior across multiple datasets provides motivation for continued investigation.

10 Potential Operational Impact

Although Reliability Validation V1 is primarily a benchmark-validation study, the observed results suggest several potentially important operational implications.

Predictive awareness can provide value beyond simple fault detection.

Early identification of degradation trajectories may allow operators to intervene before failures occur, thereby improving planning, maintenance scheduling and operational continuity.

Potential application areas include:

- aviation maintenance,
- industrial rotating machinery,
- manufacturing systems,
- energy infrastructure,
- predictive maintenance platforms,
- fleet health monitoring,
- critical infrastructure operations.

The principal value of predictive warning is not merely detecting faults.

Its value lies in providing time.

Additional warning horizon may allow maintenance planning, operational optimization and risk mitigation before degradation progresses into failure.

As industrial systems become increasingly interconnected and economically important, predictive awareness may become a significant component of future operational resilience strategies.

Reliability Validation V1 therefore contributes not only to predictive-maintenance research but also to broader investigations of infrastructure resilience and continuity.

Illustrative Operational Impact Assessment

Table 5: Illustrative Operational Impact Assessment

Area	Technical Effect	Potential Operational Benefit
Predictive Maintenance	Earlier degradation awareness	Improved maintenance planning
Industrial Equipment	Early-warning generation	Reduced unplanned downtime
Fleet Monitoring	Degradation tracking	Improved asset utilization
Critical Infrastructure	Failure-risk awareness	Increased operational resilience
Aviation Systems	Predictive health monitoring	Enhanced maintenance scheduling

These estimates are presented as engineering interpretations of validation results rather than validated economic forecasts.

Quantitative economic conclusions will require future pilot studies and operational deployments.

11 Critical Infrastructure Perspective

Reliability engineering is increasingly interconnected with broader critical-infrastructure systems.

Failures occurring within industrial equipment, transportation systems, energy systems and communications infrastructure can propagate across organizational and technological boundaries.

For this reason, predictive maintenance can no longer be viewed solely as an equipment-level problem.

It is increasingly becoming a resilience problem.

Within the GSC-X validation ecosystem, Reliability Sandbox complements Battery Sandbox and Energy Network Sandbox.

Each branch addresses a different aspect of infrastructure stability:

- Reliability Sandbox focuses on degradation and predictive maintenance.
- Battery Sandbox focuses on energy-storage health and end-of-life prediction.
- Energy Network Sandbox focuses on cascade suppression and network resilience.

Together these branches suggest the possibility of a broader predictive-resilience framework.

Rather than treating reliability, energy storage and network stability as completely separate domains, future infrastructure systems may benefit from shared executive-state awareness mechanisms.

The objective is not to replace existing monitoring systems.

Instead, executive-state architectures may function as complementary layers capable of extending predictive awareness and operational resilience.

Although substantial validation and pilot work remain necessary, the current results provide encouraging evidence that predictive-resilience architectures may represent a useful direction for future infrastructure technologies.

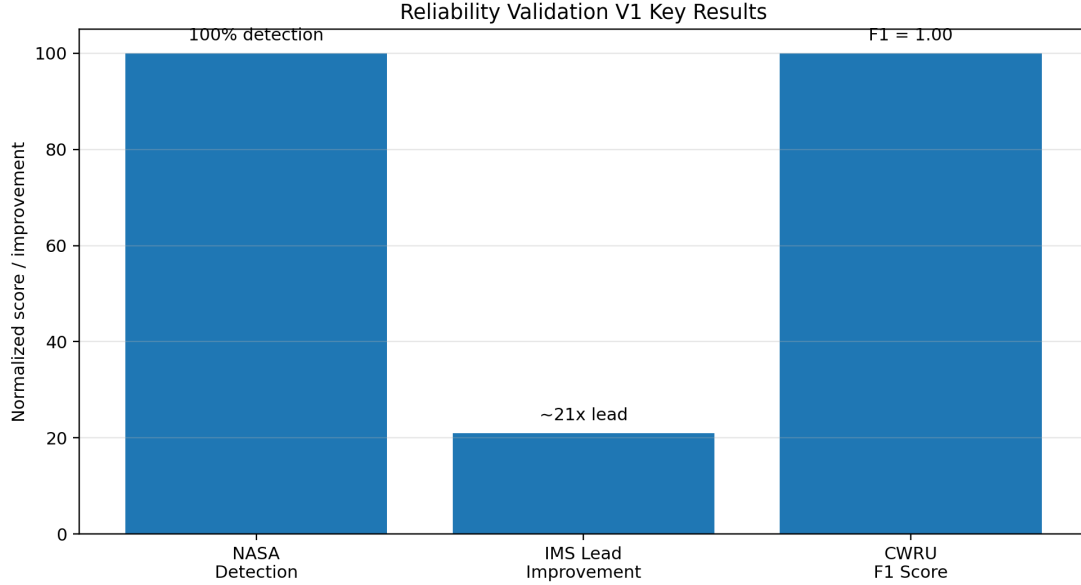


Figure 4: Summary of the principal Reliability Validation V1 results. The validation campaign achieved complete degradation detection across NASA CMAPSS, approximately twenty-one-fold lead-horizon improvement on IMS Bearings and perfect fault-state separation on CWRU Robust V2.

12 Limitations

Several important limitations should be acknowledged.

First, all validation results presented in Reliability Validation V1 were obtained using public benchmark datasets.

Although NASA CMAPSS, IMS Bearings and CWRU are widely accepted validation environments, they do not fully replicate real operational deployments.

Second, no Hardware-in-the-Loop (HIL) experiments have yet been performed within the Reliability Sandbox.

Consequently, real-time integration behavior remains outside the scope of the present study.

Third, the current validation campaign does not include direct integration with commercial predictive-maintenance platforms, industrial SCADA systems or enterprise asset-management systems.

The presented results should therefore be interpreted as validation evidence demonstrating predictive-awareness capabilities rather than proof of operational deployment performance.

Finally, the executive-state variables investigated in this work remain part of an evolving research framework.

Additional independent validation, external reproduction and industrial pilot studies will be necessary to fully evaluate long-term operational effectiveness.

Despite these limitations, the diversity of datasets included in Reliability Validation V1 provides encouraging evidence that executive-state architectures can maintain useful predictive behavior across multiple reliability domains.

13 Conclusion

Reliability Validation V1 demonstrates that executive-state architectures based on Context Memory (CM), Early Limit Indicator (ELI) and Predictive Transition Memory (PTM) can provide meaningful predictive awareness across multiple reliability benchmarks.

The validation campaign included NASA CMAPSS FD001–FD004, IMS Bearings and CWRU Robust V2 datasets.

Together these datasets span turbofan-engine degradation, bearing wear, vibration telemetry and machine-fault diagnostics.

Across NASA CMAPSS, the validation covered 709 engines and achieved complete degradation detection coverage.

The strongest executive-state configuration produced a substantial improvement in predictive-warning quality compared to baseline monitoring approaches.

Additional validation using IMS Bearings demonstrated significant lead-horizon improvements, while CWRU Robust V2 achieved perfect fault-state separation.

The broader significance of these results extends beyond predictive maintenance alone.

When considered together with Battery Validation V1 and Energy Validation V1, Reliability Validation V1 contributes to an emerging predictive-resilience framework focused on asset health, energy storage systems and infrastructure continuity.

The presented results do not establish a universal predictive-maintenance law.

However, they provide encouraging evidence that common executive-state principles may remain useful across multiple classes of infrastructure systems.

Reliability Validation V1 therefore represents both a predictive-maintenance validation study and a contribution toward the broader objective of developing predictive-resilience technologies for future industrial and critical-infrastructure environments.

From a wider perspective, Reliability Validation V1 strengthens the growing GSC-X validation ecosystem by providing reproducible evidence across multiple independent datasets and machine classes.

The consistency of observed results supports continued investigation into executive-state architectures as a potential complement to conventional monitoring and maintenance systems.

14 Future Work

Future development of the Reliability Sandbox will focus on:

- expanded NASA CMAPSS studies,
- additional IMS validation campaigns,
- additional CWRU robustness studies,
- predictive-maintenance benchmark expansion,
- Hardware-in-the-Loop validation,
- digital-twin integration,
- SCADA integration studies,
- enterprise asset-management integration,
- industrial pilot projects,
- cross-validation with Battery Sandbox,
- cross-validation with Energy Network Sandbox,
- predictive-resilience architecture development.

A long-term objective is the investigation of whether executive-state architectures can function as complementary predictive-awareness layers across multiple classes of industrial and infrastructure systems.

The goal is not to replace existing monitoring technologies.

Instead, future work will evaluate how executive-state variables can extend operational awareness, predictive warning capability and resilience-oriented decision support.

Particular emphasis will be placed on integration with real-world infrastructure environments where predictive awareness may contribute to reliability, safety and operational continuity.

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All scientific conclusions, interpretations and responsibility for the contents of this publication remain solely with the author.

Reproducibility and Public Repository

Validation artifacts, benchmark summaries, technical reports and public project documentation are available at:

<https://github.com/sisacek66198068/GSC-X>

Interactive Demonstrator

The interactive Reliability Sandbox presenting validation results, benchmark summaries and predictive-maintenance demonstrations described in this study is available at:

<https://reliability.gds-gsc-x.com>

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